

style variations in data and models
what about benchmarks? evaluating by preferences, style matters
References, complex knowledge, and alignment.
Knowledge extraction and alignment

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takeaways:
- ① You might not want to fine-tune on real knowledge, even that's the LLM use cases
 - ② In principle, 'RL' style correctness feedback could help.
 - ③ Knowledge storage and extraction in LMs is messy, and nuanced.

Safety: safety controls are needed for LLM deployed in public end-users.

(*) Safety-tuning: post-training / SFT instruction tuning can dramatically change the model safety profiles.

For SFT data:

- ①: instruction fine-tuning (SFT) works best when we are just extracting pre-training behaviors, not adding new data for pre-tuning
- ②: Adding (factually correct) data can sometimes hurt
- ③: Small amounts of the right kinds behavior (safety, instruction-following, etc) make a big difference, but there is a long-tail that benefits from more data.

How to do instruction-base fine-tuning?

- ①: turn instruction-tuning into pre-training

e.g. $\Rightarrow$ pre-train on web/pre-training data
 $\Rightarrow$ Mix in instruction-tuning data into pre-training
 $\Rightarrow$ Do an actual (but short) instruction-tuning around

mllm, jetmllm models: two-stage training

- ① Data mixture of stable stage:
common crawl, 170ma, C4, p1e...
- ② Data mixture of decay stage:
same from above, code-SFT, ~~data~~
knowledge-SFT,
Math-SFT
logic-SFT ~~data~~

Step #2 From imitation to Optimization.

imitation: SFT $\rightarrow$ pure generative modeling perspective

$\rightarrow$ Requires samples from Reference policy

Optimization (RLHF) $\rightarrow$ Maximize some Reward function that we measure

$\downarrow$
why optimize? G-V gap?

$\downarrow$ LLMs are policies, not a model of some distribution

$\downarrow$ standard RLHF / data set up: pairwise feedback setup

$\downarrow$ annotator / labeler for RLHF data: crowdsourcing ethics